

Modal parameter identification and damage detection of glass curtain walls using 3D-DIC and video motion magnification

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ABSTRACT

Modal parameters are critical indicators for damage assessment in structural health monitoring (SHM). For glass curtain wall systems, the extraction of modal shapes remains a significant technical challenge. Consequently, most existing methods for structural adhesive damage detection predominantly rely on the variations of the modal frequency, which however, maybe inadequate for precise defect identification and localization despite their correlation with damage severity. To address these limitations, this study presents a novel method for extracting modal shapes of glass curtain walls based on three-dimensional digital image correlation (3D-DIC) and video motion magnification (VMM) techniques. The effectiveness of the proposed approach was validated through indoor experiments on glass curtain wall specimens with various levels of structural adhesive damage. Furthermore, a method for identifying and localizing structural adhesive damage was developed based on the first-order modal shape information of the glass curtain walls. The results demonstrated that the proposed approach, combining 3D-DIC and VMM, can accurately extract low-order modal parameters of glass curtain walls. By taking the undamaged condition as a baseline, structural adhesive damage can be precisely identified and localized through correlation analysis of modal shape components at boundary nodes.

1. Introduction

Glass curtain walls have become a prevalent choice for the facades of high-rise buildings due to their modern aesthetic appearance and excellent performance [1,2]. A typical glass curtain wall system consists of glass panels, supporting frames, and structural adhesives that bond these components together. The structural adhesive plays a critical role in transferring external loads acting on the glass panels to the main building structure, thereby maintaining the overall stability and integrity of the curtain wall system [3,4].

However, under the continuous influence of external loads and environmental factors, the bonding strength of the structural adhesive tends to gradually degrade, leading to issues such as debonding or adhesive failure. Such deterioration compromises the structural stability of the glass curtain wall system, posing significant risks to public safety and property [5,6]. Therefore, accurate structural assessment of existing curtain walls is critically important to prevent potential incidents such as glass panel detachment.

Damage detection methods for structural adhesives can generally be

categorized into static and dynamic approaches. Static detection methods primarily evaluate the working condition of structural adhesives through material property testing, with commonly used techniques including the suction cup and airbag method [7] and the cutting and drawing method [8]. Although static methods can provide accurate results, they are often limited to sampling inspections due to their destructive nature and operational complexity. As a result, the need for non-destructive and efficient damage detection techniques has driven the development of dynamic detection methods.

The modal parameters of a structure characterize its inherent properties and are solely related to the material properties and boundary conditions of the structure [9,10]. As such, they are often employed for damage identification in structures. For instance, Wang et al. proposed an interval damage identification method based on a fused indicator combining strain mode shapes and natural frequencies [11,12]. Shi et al. addressing uncertainties encountered in the damage identification process, introduced a damage identification strategy based on probability regularization [13,14]. For glass curtain wall systems, damage to the structural adhesive implies a change in the boundary conditions of the

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glass panels [15]. Therefore, the modal parameters of the glass panels can be utilized as indicators for identifying structural adhesive damage.

However, due to the lack of effective methods for extracting modal shapes, current dynamic detection techniques for structural adhesive damage primarily rely on natural frequency information of the curtain wall. For instance, Pan et al. installed accelerometers on glass panels to collect acceleration response data and successfully identified structural adhesive damage based on differences in driving-point accelerations before and after the damage occurred [16,17]. Huang et al. used a laser Doppler vibrometer to remotely measure the vibration response of glass panels and found that the modal frequencies of the panels gradually decreased as the extent of structural adhesive damage increased [18–20].

Although changes in the modal frequencies of the glass panels can reflect the severity of structural adhesive damage, these frequency-based indicators are quantitative and cannot intuitively describe the overall damage condition or accurately localize the failure regions. In contrast, modal shapes correspond to the vibration patterns of the structure, and variations in boundary conditions are often directly manifested in these shapes. Hence, using modal shapes for structural adhesive damage identification can potentially yield more accurate results [21,22].

Currently, the acquisition of modal shapes for glass curtain walls primarily relies on experimental modal analysis. For example, Jiang et al. collected vibration signals by installing accelerometers on the surface of glass curtain walls and calculated frequency response functions to obtain modal frequencies and modal shapes [23]. However, densely distributed sensors may affect the mechanical properties of the glass panels. To address this limitation, this study employs three-dimensional digital image correlation (3D-DIC) to acquire vibration data of glass curtain walls. In addition, due to the high stiffness of glass panels, their vibration amplitudes under external excitation are relatively small and often contain multiple coupled modal components. Therefore, extracting modal shapes requires the separation and amplification of single-mode vibration components. To achieve this, video motion magnification (VMM) is introduced to process vibration videos

of the glass curtain walls.

Based on these considerations, this study integrates 3D-DIC and VMM technologies to propose an efficient and stable method for extracting modal shapes of glass curtain walls. By analyzing variations in modal shape components at the boundaries of the curtain wall under different working conditions, a localization strategy for structural adhesive damage is introduced. The structural layout of this paper is organized as follows: Section 2 introduces the fundamental principles of 3D-DIC and VMM, as well as the workflow for extracting modal parameters of glass curtain walls through their combined application. Section 3 describes the design of the indoor experiments and the development of the finite element model, followed by a detailed analysis and discussion of both the experimental and numerical results from the perspectives of displacement, frequency, and modal shape. Section 4 explores a damage identification strategy for structural adhesives based on modal shape information and establishes a series of experimental conditions to validate the effectiveness and general applicability of the proposed method. Section 5 summarizes the main conclusions and outlines potential directions for future research.

2. Study methodology

2.1. Work plan – modal parameters extraction and damage identification

As illustrated in the flow chart shown in Fig. 1, the adapted work plan and research methodology for extracting the modal parameters of glass curtain walls using 3D-DIC and VMM techniques – including the identification and localization of structural adhesive damage – consist of the following five procedural steps:

Step (a). A 3D-DIC system is established using a binocular camera and a data processing terminal to capture vibration videos of the glass curtain wall panels under external excitation. Preprocessing operations, such as grayscale conversion, are performed on the video frames, and camera calibration is conducted to obtain the intrinsic and extrinsic parameters of the camera.

Step (b). The 3D-DIC program is then employed to measure the

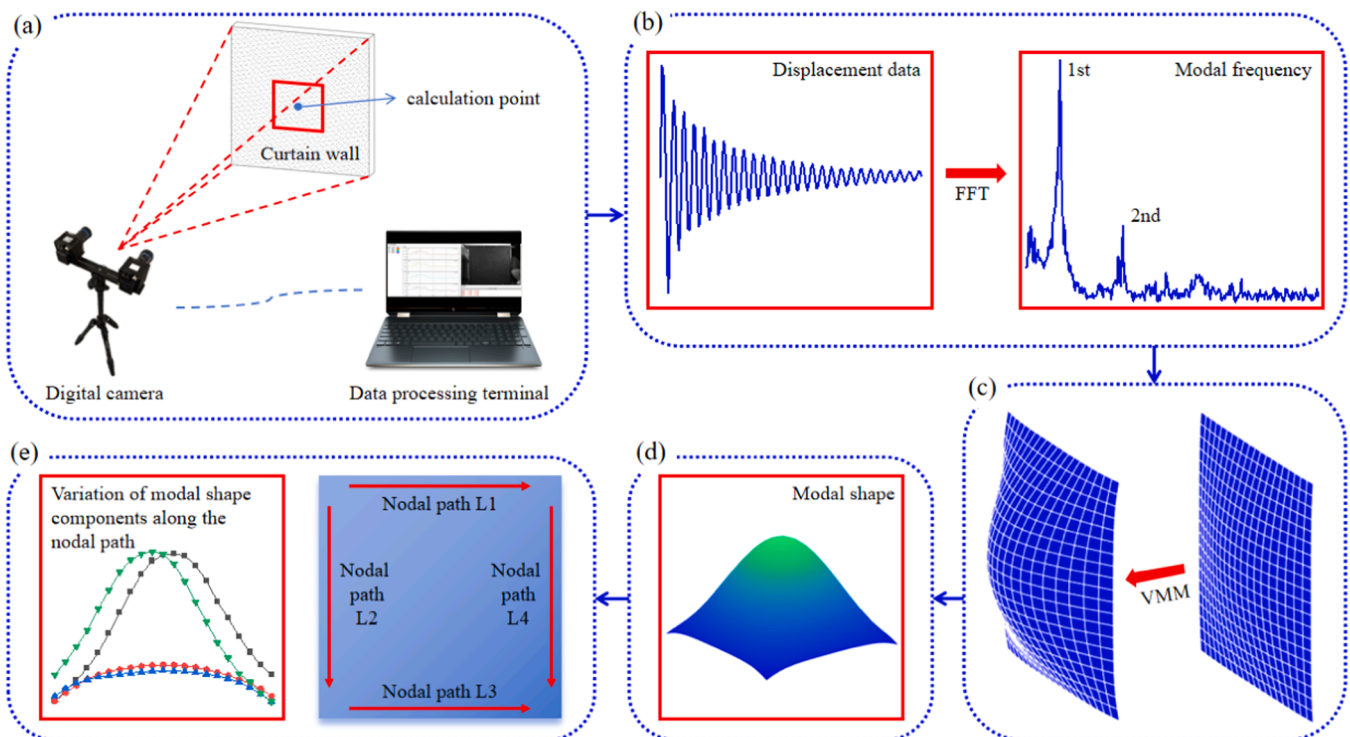


Fig. 1. Work plan flow chart – Image acquisition, data processing, and damage identification.

displacement time-history response curves of the calculation points. By performing a Fast Fourier Transform (FFT) on the displacement data, the corresponding frequency spectrum is obtained, from which the modal frequencies of the curtain wall structure are determined.

Step (c). Based on the modal frequencies of the glass panels, the frequency band interval for the band-pass filter in the VMM algorithm is defined, and an appropriate motion magnification factor is selected. After inputting the filter band and magnification coefficient, the original videos captured by the binocular camera are processed separately for motion magnification.

Step (d). The displacement data of calculation points are extracted from the VMM-processed videos using 3D-DIC, and the time instances corresponding to the displacement extrema are identified. Simultaneously, stereo matching and three-dimensional reconstruction operations are performed to obtain the 3D morphology of the glass panel at these specific moments. After data normalization and surface fitting, the modal shapes of the curtain wall structure are derived.

Step (e). Finally, the identification and localization of structural adhesive damage are performed based on the differences in modal shapes of the glass panels under various working conditions. Four boundary nodal paths are first defined. The modal shape components along these paths under the undamaged condition are established as baseline values. For any given condition, the correlation coefficients between its modal shape components and the baseline values are calculated. When a correlation coefficient falls below a predefined threshold, structural adhesive damage is identified along the corresponding boundary.

Compared with traditional modal parameter extraction methods and structural adhesive damage identification strategies, the proposed approach offers two main advantages. First, it replaces data acquisition devices such as accelerometers and displacement sensors with industrial cameras, simplifying the data collection process and avoiding the impact of sensor installation on the inherent vibration characteristics of the structure. Second, it reduces the cumbersome process of modal shape computation to a straightforward digital image processing task. By integrating 3D-DIC and VMM technologies, the modal shapes of glass curtain walls can be directly extracted, enabling intuitive identification and precise localization of structural adhesive damage.

2.2. Three-dimensional digital image correlation

The DIC technique is a non-contact, full-field optical measurement method. This approach requires relatively low hardware specifications whilst offering high measurement accuracy and computational efficiency, making it widely applicable across various fields [24,25]. The fundamental principle of the DIC method is illustrated in Fig. 2. A calculation point P is selected in the reference image, and a local pixel region centered around this point is defined as the reference subset. Through correlation analysis, the corresponding subset that exhibits the highest correlation with the reference subset is identified in the

deformed image, thereby determining the position P' of the calculation point in the deformed image [26].

In practical measurements, structural deformation often occurs, leading to significant disparities between the reference and deformed images. Therefore, a shape function is employed to accurately characterize the actual configuration of the corresponding subset [27]. Given the pronounced non-uniform deformation observed in glass panels during vibration, a second-order shape function is adopted in this study [28].

Furthermore, to enhance the computational accuracy of the DIC method, the search process for the corresponding subset is generally divided into integer-pixel matching and sub-pixel matching. Since the result of integer-pixel matching is typically used as the initial value for sub-pixel iteration, a highly accurate correlation criterion is required. Considering factors such as uneven illumination and time-varying light intensity in practical measurement environments, this study adopts the Zero-Normalized Cross-Correlation (ZNCC) function, which exhibits strong resistance to such interference [29]. Meanwhile, the sub-pixel matching process involves extensive iterative computations. To improve algorithmic efficiency while maintaining computational precision, the Inverse Compositional Gauss-Newton (IC-GN) algorithm is selected in this work [30].

3D-DIC extends conventional DIC by integrating it with binocular stereo vision, enabling the measurement of three-dimensional structural deformations. Compared with 2D-DIC, it offers greater flexibility in camera setup, making it more suitable for practical engineering measurements [31]. The workflow of the 3D-DIC method employed in this study is illustrated in Fig. 3. First, stereo matching is performed to obtain the corresponding coordinates of calculation points in the left and right camera images. DIC technology is then applied to track the pixel coordinate variations of these calculation points throughout the image sequence. Next, by incorporating camera parameters obtained from calibration, the spatial coordinates of the calculation points are determined. Finally, the displacement data in the corresponding directions are derived based on changes in the spatial coordinates of the calculation points. For further theoretical details on the principles of 3D-DIC, readers may refer to [32].

2.3. Video motion magnification

Depending on how the target motion is described and processed, VMM techniques can be categorized into Lagrangian-based VMM [33] and Eulerian-based VMM [34,35]. The latter does not require explicit estimation or tracking of the motion trajectories of the image feature points or patches, offering higher computational efficiency and stronger robustness – making it the mainstream approach at present.

Eulerian-based VMM can be further divided into linear Eulerian motion magnification [34] and phase-based motion magnification (PBMM) [35]. The PBMM algorithm leverages the time-shift property of the Fourier transform, converting spatial changes in the image domain into phase changes in the frequency domain for indirect processing. This approach translates noise without amplifying it, achieving ideal magnification effects. On this basis, this study employed the PBMM algorithm to magnify the vibrations of the glass curtain wall.

The fundamental principle of the PBMM algorithm is illustrated in Fig. 4. After inputting the original video, each frame is decomposed using a complex steerable pyramid to obtain a series of local amplitude spectra and local phase spectra at different scales and orientations. The phase difference data are then calculated based on the obtained local phase spectra. Next, a frequency band interval is specified for temporal band-pass filtering to extract phase differences within the desired frequency range. A suitable magnification factor is applied, and the filtered phase difference signal is multiplied by this factor to achieve a linearly magnified result. Finally, the magnified data are reconstructed through an inverse steerable pyramid and combined with the high-pass and low-pass residuals from the forward decomposition process to generate the

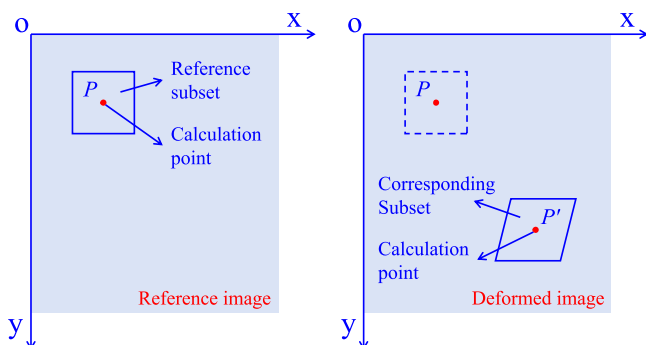


Fig. 2. Schematic illustration of DIC principle.

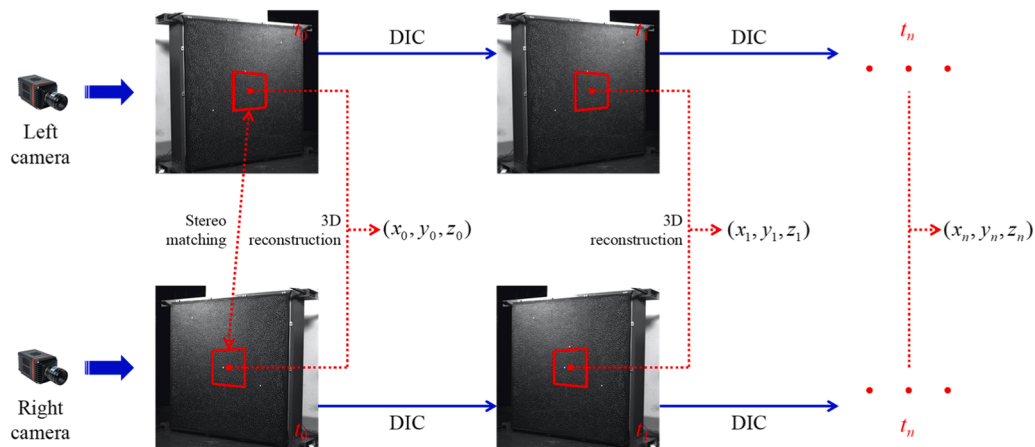


Fig. 3. Workflow plan of the 3D-DIC method.

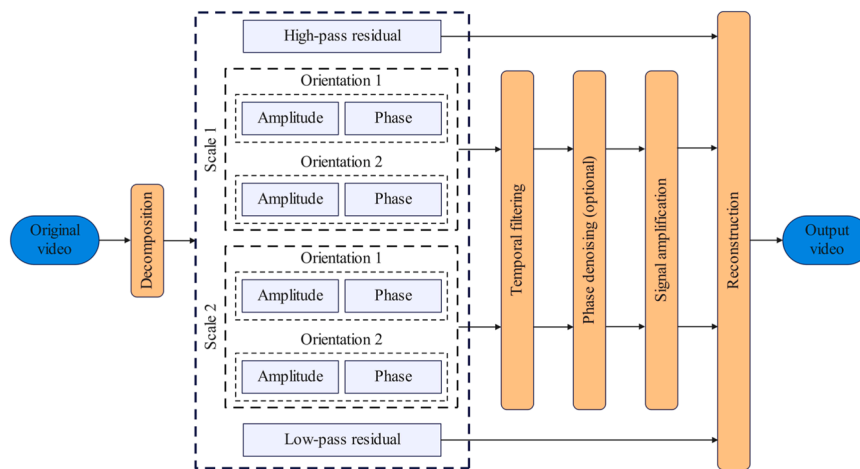


Fig. 4. Workflow plan of the PBMM algorithm.

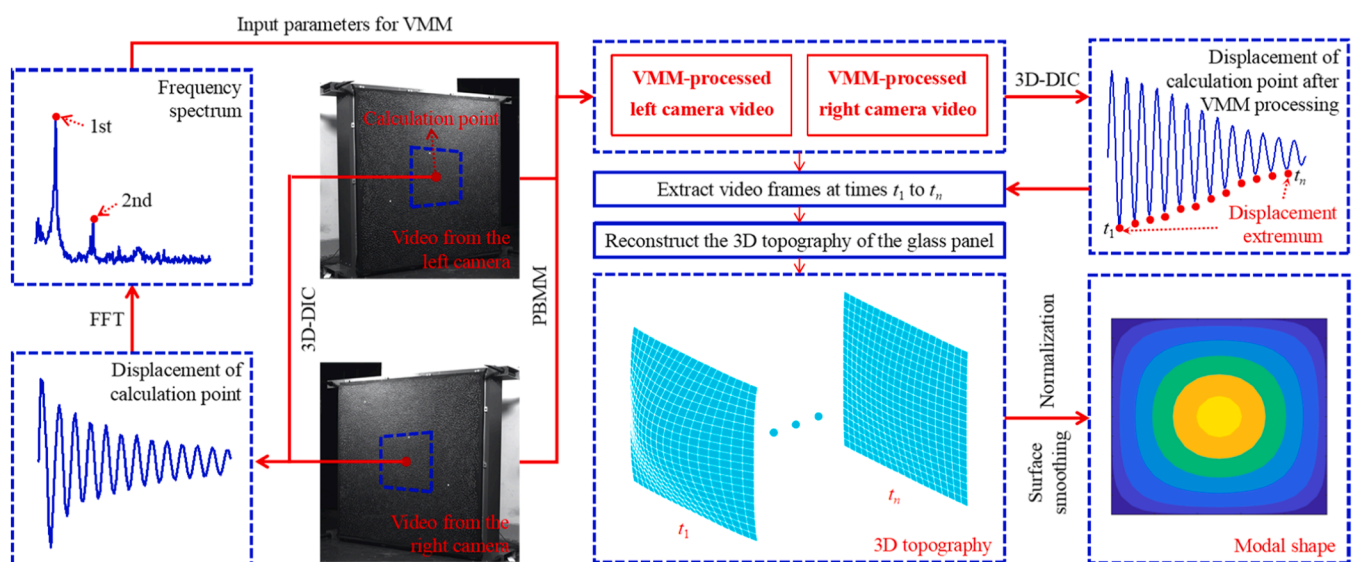


Fig. 5. Workflow for extracting modal parameters of glass curtain walls using the integrated 3D-DIC and VMM approach.

motion-magnified output video. For the detailed derivation process of the PMM algorithm, please refer to the literature [35].

2.4. Modal parameters extraction

The workflow for extracting the modal parameters of glass curtain walls based on the combined application of 3D-DIC and VMM techniques is illustrated in Fig. 5. After acquiring the original vibration video of the curtain wall structure, the center point of the glass panel is selected as the calculation point, as shown in Fig. 5. The displacement time-history curve of this calculation point is obtained using the three-dimensional DIC method, and a Fast Fourier Transform (FFT) is then performed to identify the modal frequencies of the glass panel. Based on the identified modal frequencies, the frequency band intervals for band-pass filtering in the PBMM algorithm are determined.

It should be noted that the full-field vibration response of the glass panel represents a superposition of multiple modal shapes. Separating the vibration components corresponding to individual modes from the full-field response is therefore a prerequisite for extracting single-mode shapes. During band-pass filtering, a narrow frequency band centered on each modal frequency is employed. For example, with a first-order modal frequency f_{1st} , if the bandwidth is set to 0.5 Hz, the frequency band range is $(f_{1st} - 0.25, f_{1st} + 0.25)$ Hz.

Next, after setting parameters such as the filter range and magnification factor, the PBMM algorithm is applied to magnify the vibration of the glass panel in the original video. The displacement time-history curve of the calculation point is then re-obtained using the 3D-DIC method, and the time instants (t_1 to t_n) corresponding to the displacement extrema are identified. Stereo matching and three-dimensional reconstruction are subsequently performed to acquire the three-dimensional morphology of the glass panel at each time instant (t_1 to t_n). Finally, the reconstructed three-dimensional morphology data at all time instants are normalized and averaged to derive the corresponding modal shape, as exemplified in Fig. 5. To further enhance the quality of the modal shape, surface smoothing can be applied to reduce noise points.

3. Laboratory experiment and finite element analysis

3.1. Experimental steps and finite element model

The overall layout of the indoor experimental setup is shown in Fig. 6. Video acquisition was performed using two high-speed industrial cameras of the same model. The cameras have a resolution of 1936×1464 pixels, an effective pixel count of 2.8 million, a pixel size of $4.5 \mu\text{m}$, and a frame rate of 400 FPS.

As illustrated in Fig. 6, the experimental system mainly consisted of a glass curtain wall structure and a supporting frame. The glass curtain wall structure comprised a single-layer tempered glass panel and a metal frame. The glass panel measured $1000 \text{ mm} \times 1000 \text{ mm} \times 5 \text{ mm}$ and was bonded to the metal frame using silicone structural adhesive (width:

10 mm, thickness: 5 mm). The metal frame was anchored to the supporting system with metal clamps. The supporting system was assembled from aluminum alloy components and threaded anchor rods, which provided structural support and ensured load transfer. One end of each threaded anchor rod was fixed to the wall to maintain the stability of the aluminum components during excitation.

Since this study focuses on methodological development, the speckle pattern on the glass panel surface was manually applied. To facilitate practical engineering applications, further research on speckle application techniques is needed in future work. The virtual speckle projection technique described in Ref. [36], as well as the combined use of a polarization RGB camera and fluorescent speckles presented in Ref. [37], both provide potential solutions for applying speckle patterns to glass curtain walls without compromising their visual appearance.

Two test points were established in this study (as shown in Fig. 6(d)), with miniature accelerometers and laser displacement sensors (LDS) installed at all measurement locations (see Fig. 6(c)). The accelerometer model used was DH5922N, and the laser displacement sensor was KEYENCE LK-G5001. External excitation was applied by striking the center of the glass panel with an impact hammer.

To validate the effectiveness of the proposed modal shape extraction method for glass curtain walls, three working conditions were designed, as illustrated in Fig. 7. In Condition 1, the structural adhesive remained intact. Condition 2 involved adhesive damage along a single boundary, while Condition 3 involved damage along two adjacent boundaries. The damage was simulated by partially removing adhesive segments, each 50 cm in length, symmetrically distributed along the boundary center.

Furthermore, based on the actual dimensions of the experimental specimens, finite element (FE) models of the glass curtain wall structures under different working conditions were established using a numerical analysis software, as shown in Fig. 7. Both the glass panels and the supporting structures were modeled as isotropic, homogeneous, and linear elastic materials. The structural adhesive was modeled as an incompressible material, and its stress-strain relationship was obtained from planar tensile tests [38]. The glass panels, structural adhesive, and supporting structures were coupled using tie constraints. The specific material properties of each component are listed in Table 1.

3.2. Displacement and frequency test results

The out-of-plane displacements of the glass panel under three working conditions were measured using the 3D-DIC method and compared with the results obtained from the laser displacement sensor. Taking Condition 1 as an example, the time-history curves of the out-of-plane displacements at Test Points 1 and 2 are presented in Fig. 8. As shown in Fig. 8, the measurements obtained from the 3D-DIC and the laser displacement sensor exhibit strong agreement. Moreover, the root mean square errors (RMSE) between the 3D-DIC and laser displacement sensor data were 0.05562 mm at Test Point 1 and 0.03265 mm at Test Point 2, respectively, confirming the high accuracy of the 3D-DIC method in displacement measurement.

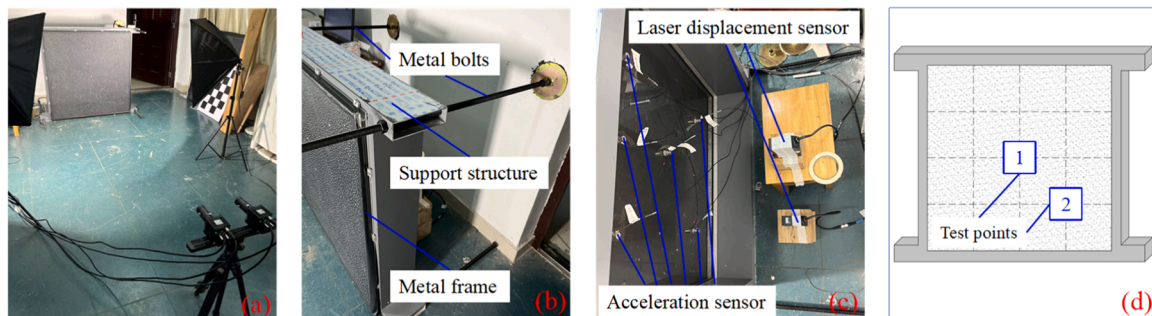


Fig. 6. Laboratory experimentation plan.

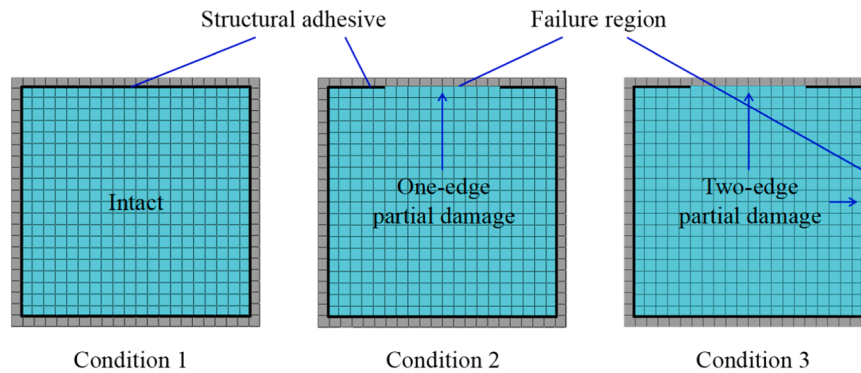


Fig. 7. Finite element models of the glass curtain wall structure under different conditions.

Table 1

Material properties.

Component	Elastic modulus/ GPa	Poisson's ratio	FE element type
Glass panel	72	0.2	S4R
Structural adhesive	0.015	0.5	C3D8I
Aluminum alloy frame	200	0.3	C3D8I

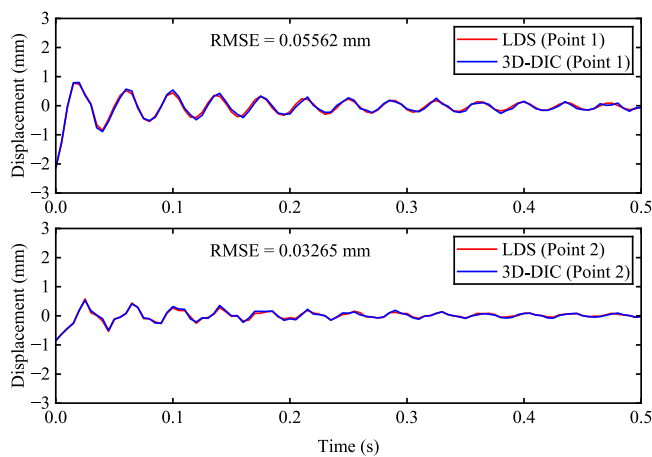


Fig. 8. Comparison of out-of-plane displacement measurement results.

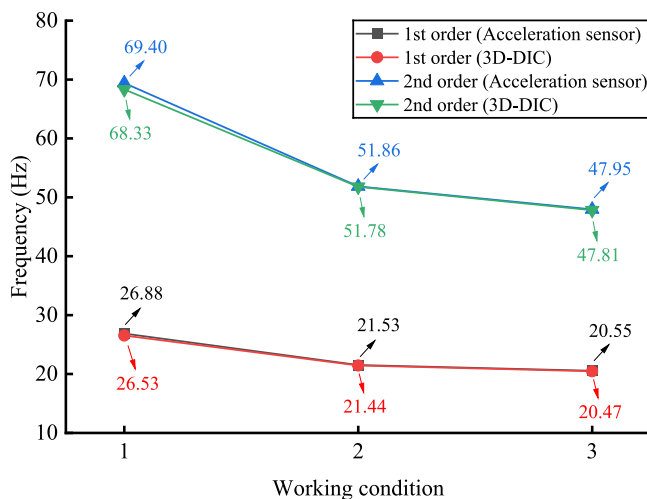


Fig. 9. Modal frequencies of the glass panel under different conditions.

The modal frequencies of the glass panel under different working conditions were obtained by performing Fast Fourier Transform (FFT) on the displacement data measured using the 3D-DIC method, as shown in Fig. 9. A comparison with the accelerometer results indicates that the modal frequencies derived from the 3D-DIC method are highly consistent with those obtained from the accelerometer, confirming the reliability of the proposed measurement approach. Furthermore, as the extent of structural adhesive damage increases, the modal frequencies of the glass panel exhibit a clear decreasing trend, which agrees well with the findings reported in Ref. [14].

3.3. Modal shape test results

The first two modal shapes of the glass panel under different working conditions, obtained through finite element analysis, are presented in Fig. 10. As observed in Fig. 10, significant alterations occur in both the first and second modal shapes following the onset of structural adhesive damage. However, the second-order modal shapes exhibit substantial variability across different conditions, making it difficult to perform a quantitative comparative analysis of the modal shape components at the nodes. In contrast, the first-order modal shape of the glass panel shows intuitive and regular variations, with differences in modal shape components mainly concentrated at the locations of structural adhesive damage. Therefore, the first-order modal shape of the glass curtain wall structure was considered more suitable for locating structural adhesive damage.

According to the modal shape extraction process shown previously in Fig. 5, the first-order modal frequencies of the glass curtain wall structure under the three working conditions were 26.53 Hz, 21.44 Hz, and 20.47 Hz, respectively. Consequently, the frequency band intervals for band-pass filtering in the VMM process were set to (26.28 – 26.78) Hz, (21.19 – 21.69) Hz, and (20.22 – 20.72) Hz, respectively.

Due to the small vibration amplitudes of the glass panels, the motion magnification factor required careful calibration. When the magnification factor is set too low, the deformation of the glass panel is insufficiently amplified, reducing the accuracy of 3D morphology reconstruction and limiting the effectiveness of modal shape extraction. Conversely, excessive magnification introduces amplified noise artifacts in the processed video frames, which distort the structural vibration patterns and compromise the modal shape reconstruction accuracy [39, 40].

Therefore, the magnification factor was gradually increased until the amplified vibration of the glass panel predominantly contained the first-order mode. Taking Working Condition 1 as an example, when the magnification factor reached 30, the time-history curve of the out-of-plane displacement at the center point of the glass panel is shown in Fig. 11(a), and its corresponding frequency spectrum obtained through FFT is presented in Fig. 11(b). As shown in Fig. 11, after magnifying the first-order vibration by 30 times, the motion of the glass panel predominantly contained only the first-order mode, while the magnified

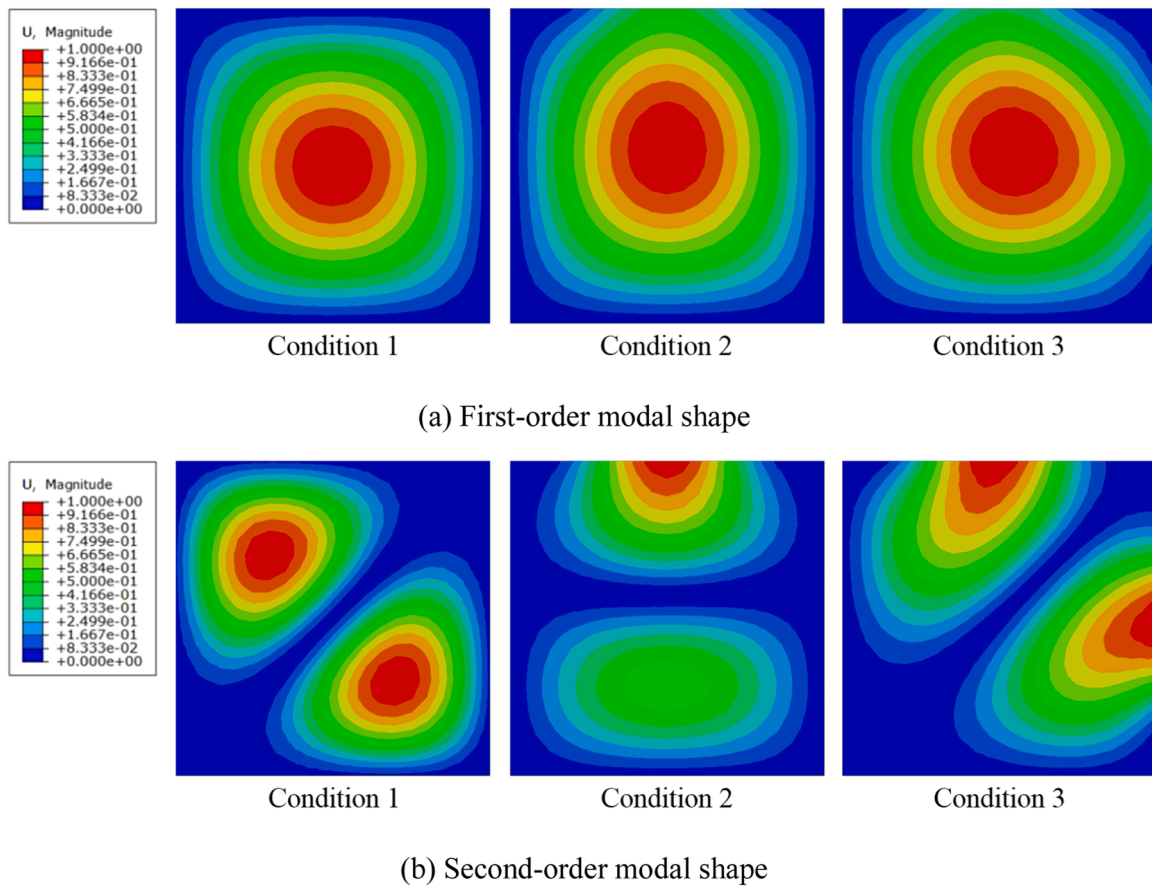


Fig. 10. Modal shapes of the glass curtain wall obtained from finite element analysis.

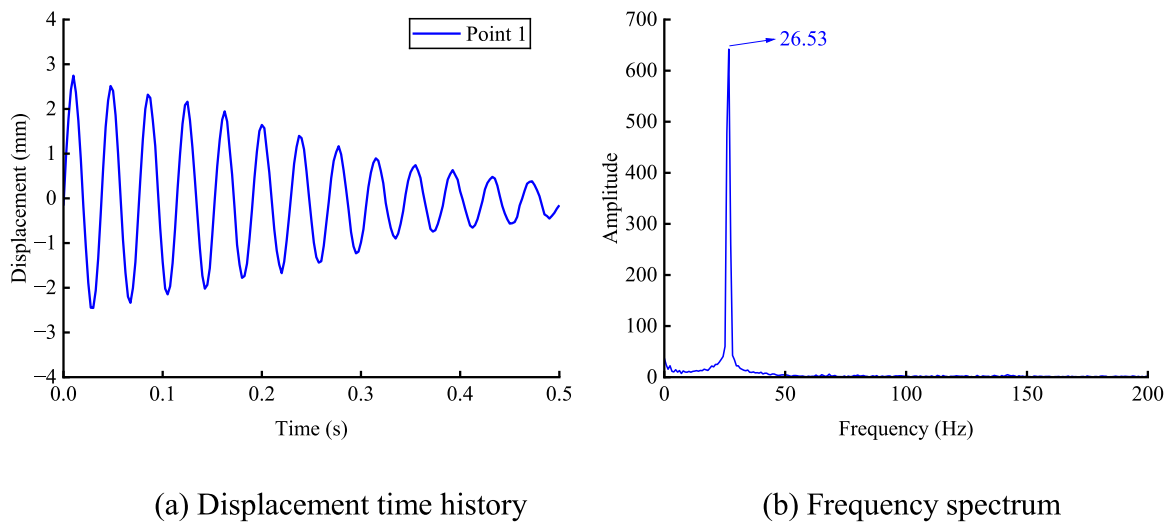


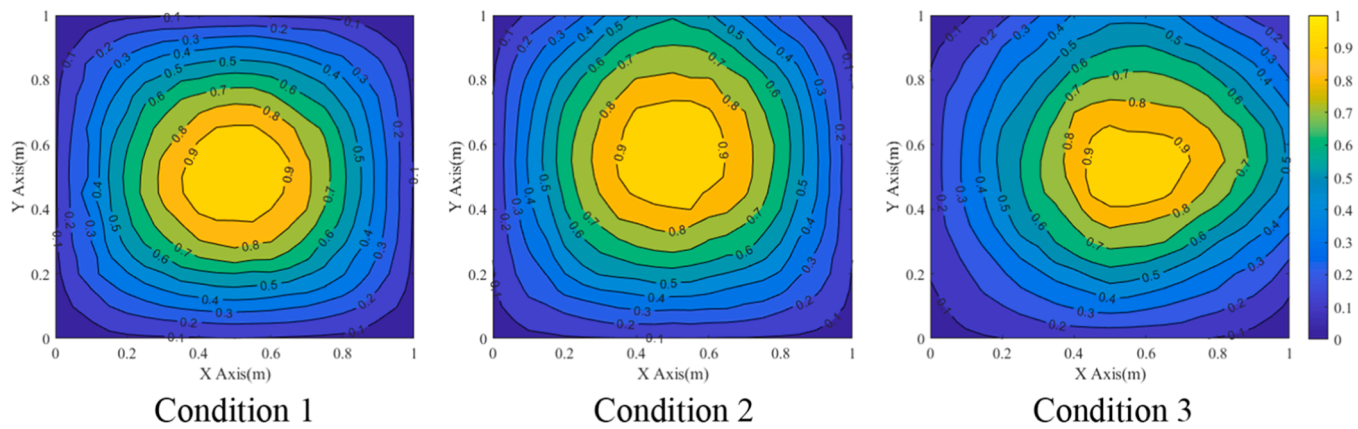
Fig. 11. Vibration response in the time and frequency domains at the selected calculation point after VMM processing.

video remained clear and free from artifact noise, making it suitable for modal shape extraction. Consequently, the magnification factor for the VMM process was ultimately determined to be 30.

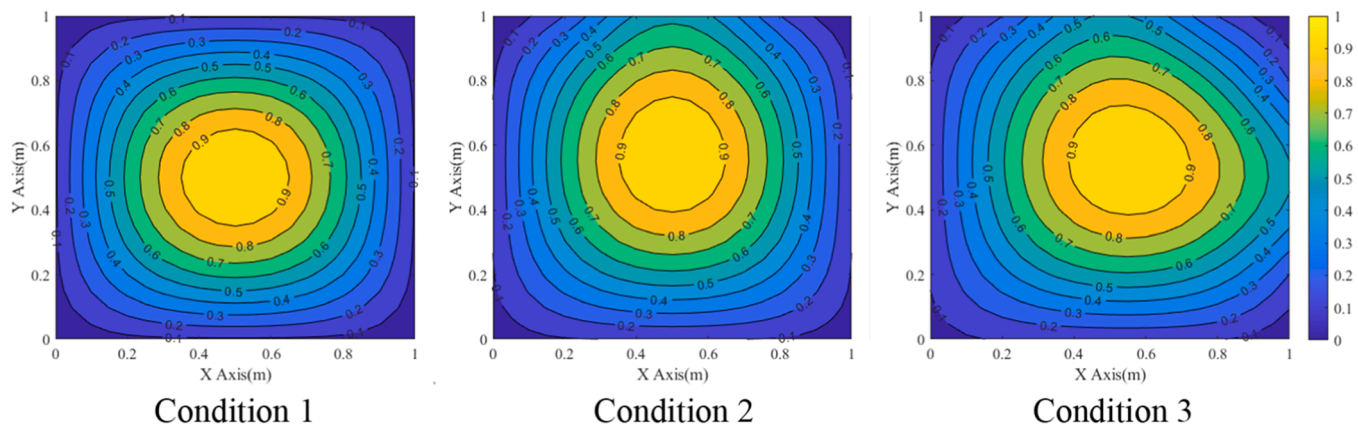
The first-order modal shapes of the glass curtain wall extracted using the combined application of 3D-DIC and VMM are shown in Fig. 12(a), while the first-order modal shapes obtained through finite element analysis are shown in Fig. 12(b).

As observed in Fig. 12, the modal shapes extracted by the two methods exhibit high overall consistency and similarity. Using the

Modal Assurance Criterion (MAC) [41] as the evaluation metric for modal shape similarity, the MAC values between the modal shapes obtained by the two methods under Conditions 1 (intact), 2 (one-edge partial debonding failure), and 3 (two-edge partial debonding failure) were calculated as 0.99928, 0.99209, and 0.99245, respectively. These results indicate that the modal shapes extracted by the two methods are highly correlated, thereby validating the effectiveness of the proposed modal shape extraction method for glass curtain walls based on 3D-DIC and VMM technologies.



(a) The combined application of 3D-DIC and VMM methods



(b) Finite element analysis

Fig. 12. Comparison of the first-order modal shapes obtained from the proposed (3D-DIC & VMM) method and finite element analysis under three conditions.

4. Structural adhesive damage identification

4.1. Modal shape-based damage identification approach

As shown in Fig. 12, significant differences can be observed in the

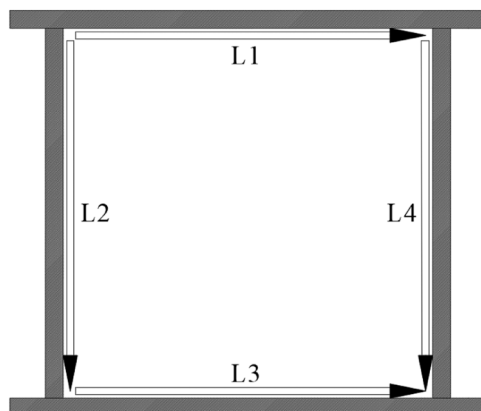


Fig. 13. Schematic plot of the preset path.

first-order modal shapes at the boundary nodes of glass curtain wall structures with varying degrees of structural adhesive damage. To visualize these variations more clearly, four nodal paths were defined, as illustrated in Fig. 13, and the variation curves of modal shape components along these paths are plotted in Fig. 14.

Fig. 14 clearly demonstrates that both the extent and location of structural adhesive damage are reflected in the variation curves of the modal shape components at the boundary nodes. In Condition 2, damage occurs along the boundary corresponding to path L1, while in Condition 3, damage is present along the boundaries corresponding to both paths L1 and L4. Compared with undamaged boundaries, the modal shape component curves along the damaged paths exhibit more pronounced fluctuations.

To standardize the procedure for assessing structural adhesive damage, the modal shape components along the four nodal paths under the undamaged condition were established as baseline values. The correlation coefficients between these baseline values and the corresponding modal shape components under the other working conditions were then calculated, as presented in Table 2.

As shown in Table 2, under Condition 2, the correlation coefficient between the modal shape components along nodal path L1 and the baseline values is only 0.88395. Under Condition 3, the correlation coefficients for paths L1 and L4 are 0.82857 and 0.79156, respectively, all

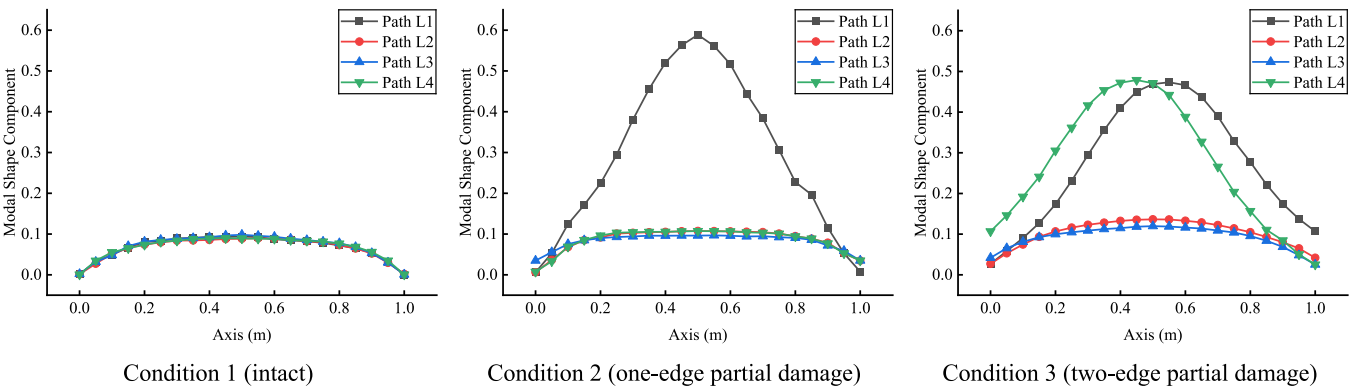


Fig. 14. Variation curves of modal shape components along the predefined nodal paths.

Table 2
Correlation coefficients of boundary modal shape components under different experimental conditions.

Correlation coefficient	Working condition	Nodal path			
		L1	L2	L3	L4
r	2	0.88395	0.98124	0.98933	0.97474
	3	0.82857	0.98010	0.9823	0.79156

Note: r denotes the Pearson correlation coefficient.

below 0.9. By contrast, the correlation coefficients for paths with intact structural adhesive all exceed 0.95. This analysis demonstrates that, during structural adhesive damage detection, a correlation coefficient threshold (e.g., 0.95) can be established. When the correlation coefficient between the modal shape components along a nodal path and the baseline values falls below this threshold, the presence of structural adhesive damage along that boundary can be inferred.

4.2. Numerical validation

To further validate the universality of the proposed structural adhesive damage identification method, finite element models were established for eight working conditions, as illustrated in Fig. 15. The first-order modal frequencies of the glass curtain wall under these eight conditions are presented in Fig. 16. As shown in Fig. 16, the frequency values for Conditions 2 and 7 are very similar. However, while damage

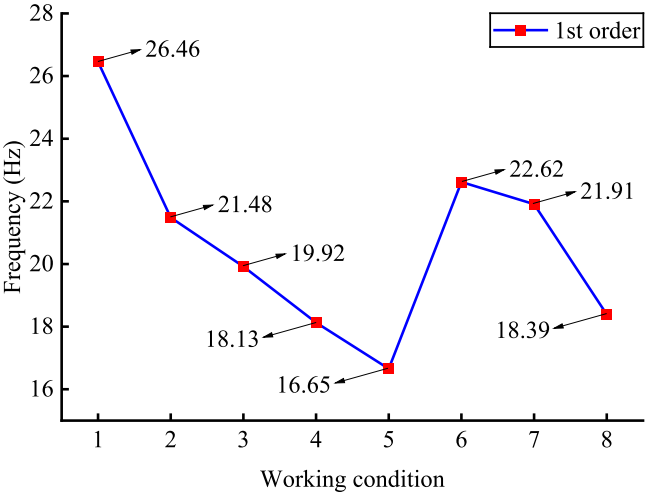


Fig. 16. Modal frequencies of the glass panel under different conditions obtained from finite element analysis.

exists only along boundary L1 in Condition 2, damage is present along both boundaries L1 and L3 in Condition 7. This observation demonstrates that using modal frequencies alone cannot effectively identify the specific damage conditions of the structural adhesive.

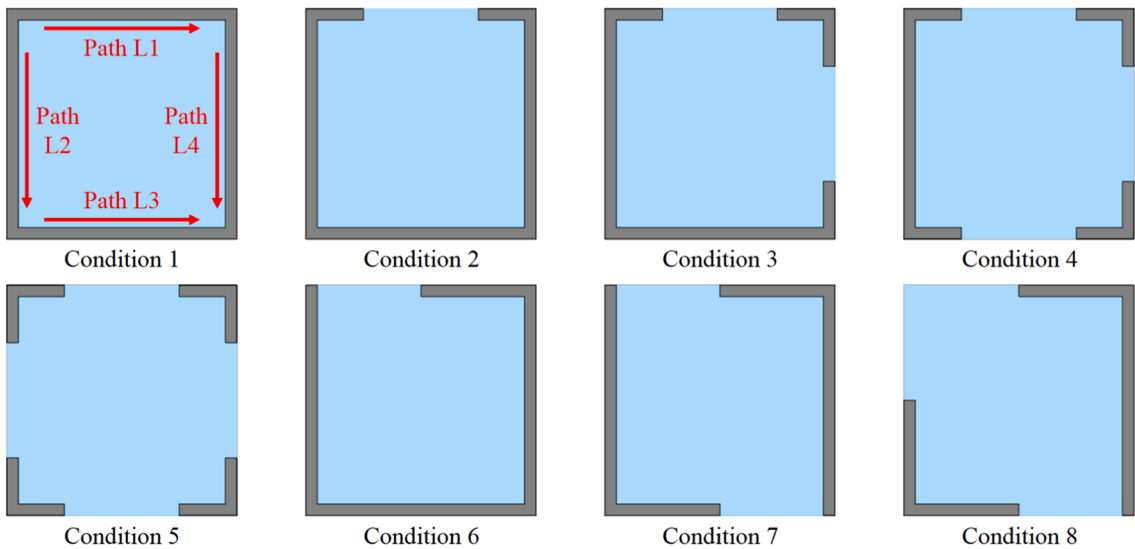


Fig. 15. Schematic illustration of the multiple adhesive degradation scenarios used in the numerical simulations.

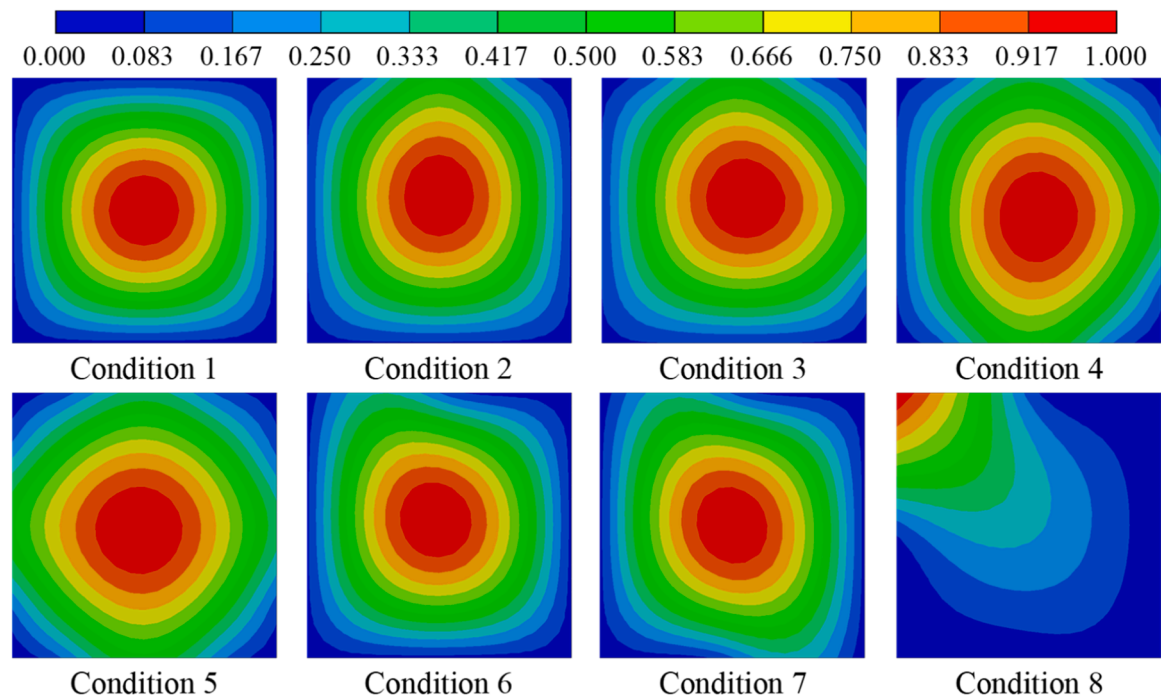


Fig. 17. First-order modal shapes of the glass panel under different conditions obtained from finite element analysis.

The first-order modal shapes of the glass curtain wall under different working conditions, obtained through finite element analysis, are shown in Fig. 17. As observed in Fig. 17, structural adhesive damage significantly affects the modal shape components at the boundary nodes. Using the modal shape components along the four nodal paths under Condition 1 (undamaged) as baseline values, the correlation coefficients between these baseline values and the modal shape components under the other working conditions were calculated and are presented in Table 3.

The results in Table 3 indicate that, regardless of the working condition, the correlation coefficients corresponding to boundaries with structural adhesive damage are all below 0.95, while those for boundaries with intact structural adhesives all exceed 0.95. These findings confirm that the proposed structural adhesive damage identification method remains effective even under complex damage scenarios, demonstrating strong universality and robustness.

5. Conclusions and recommendations

This study presented an integrated methodology for modal parameters extraction and structural adhesive damage identification in glass curtain walls by combining three-dimensional digital image correlation with phase-based video motion magnification. Laboratory experiments and finite element analyses were conducted on glass curtain wall specimens with varying adhesive conditions to validate the proposed

approach. Based on the findings, the following conclusions can be drawn:

- (1) The 3D-DIC method was successfully utilized to obtain the full-field vibration information of glass curtain walls, and combined with the VMM algorithm to extract the first-order modal shape of the glass panel. Comparison with experimental sensor measurements and finite element analysis confirmed the effectiveness of the integrated 3D-DIC and VMM approach for extracting the modal parameters of glass curtain walls.
- (2) Compared with higher-order modal shapes, the first-order modal shape of the glass panel proved to be more suitable for identifying and locating structural adhesive damage. The failure regions of the adhesive were intuitively and clearly reflected in the first-order modal shape.
- (3) A methodology for the identification and localization of structural adhesive damage in glass curtain walls was developed. The proposed approach defines nodal paths along the four boundaries of the glass panel and establishes their first-order modal shape components under undamaged conditions as baseline references. During implementation, the modal shape components obtained from test specimens are compared with these baseline values through correlation analysis. A predefined threshold of 0.95 for the correlation coefficient enables effective damage assessment—when the coefficient falls below this critical value, adhesive failure is confirmed along the corresponding boundary.

Table 3
Correlation coefficients of boundary modal shape components under different numerical simulation conditions.

Correlation coefficient	Working condition	Path			
		L1	L2	L3	L4
<i>r</i>	2	0.70048	0.97786	0.99987	0.97786
	3	0.71110	0.97946	0.97946	0.7111
	4	0.71416	0.99510	0.73721	0.72612
	5	0.72853	0.74750	0.7475	0.72853
	6	0.50848	0.98872	0.99991	0.99997
	7	0.51397	0.99022	0.51396	0.99022
	8	0.25729	0.25734	0.54846	0.99833

Note: *r* denotes the Pearson correlation coefficient.

The proposed methodology demonstrates significant advantages in detecting and localizing structural adhesive damage through its non-contact nature, operational efficiency, and quantitative evaluation capability. Unlike conventional frequency-based approaches that struggle to distinguish between different damage configurations, this method effectively identifies both the presence and specific locations of adhesive failures. Validation under multiple working conditions confirmed the method’s reliability and adaptability for practical engineering applications, providing a viable solution for the safety assessment of existing curtain wall systems.

While the research outcomes presented in this study demonstrate

strong practical applicability, the experimental methodology relied on manually applied speckle patterns on the glass panels, which presents certain limitations for real-world engineering implementation. To address this challenge, several advanced approaches have been proposed in recent studies, including digital speckle projection and the use of colorless coatings detectable only by specialized cameras. Building on these developments, future work will focus on developing non-intrusive speckle application techniques that preserve the architectural appearance of glass curtain walls, thereby enhancing both the applicability and operational convenience of the proposed damage identification methodology.

CRedit authorship contribution statement

Shuisheng Li: Writing – review & editing, Visualization, Investigation, Funding acquisition. **Wei He:** Writing – review & editing, Validation, Software, Formal analysis. **Banfu Yan:** Writing – review & editing, Validation, Project administration, Methodology, Funding acquisition. **Lei Luo:** Writing – review & editing, Writing – original draft, Software, Methodology, Investigation.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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